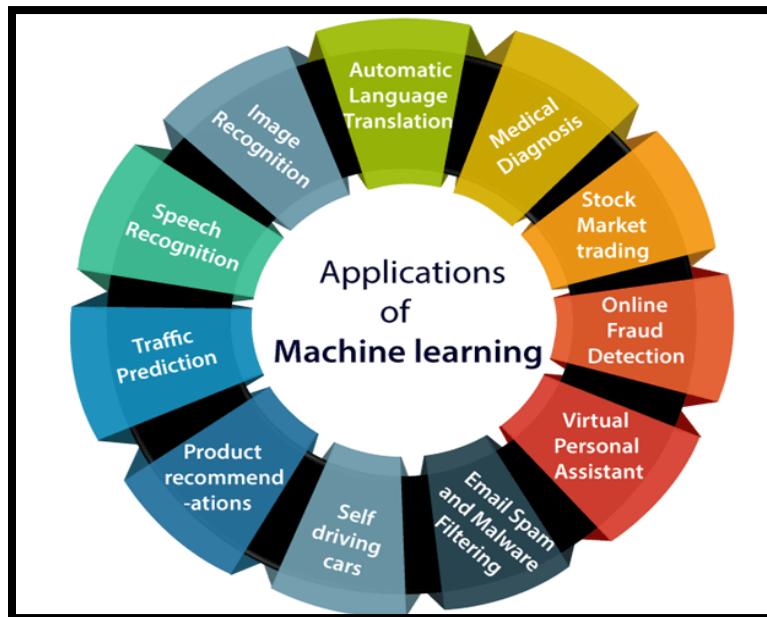


SUPERVISED LEARNING

**Explaining supervised learning algorithms, including
Regression and Classification**

INTRODUCTION

AI driven innovations have completely transformed our lives by impacting almost every industry. Technological advancements in healthcare, businesses, education, entertainment industries now extensively depend on AI. From personalised recommendations to disease detection to self-driving cars , Machine Learning algorithms are at the heart of these innovations. In machine learning, supervised learning is one such paradigm that takes insights from labelled datasets and produces predictions.



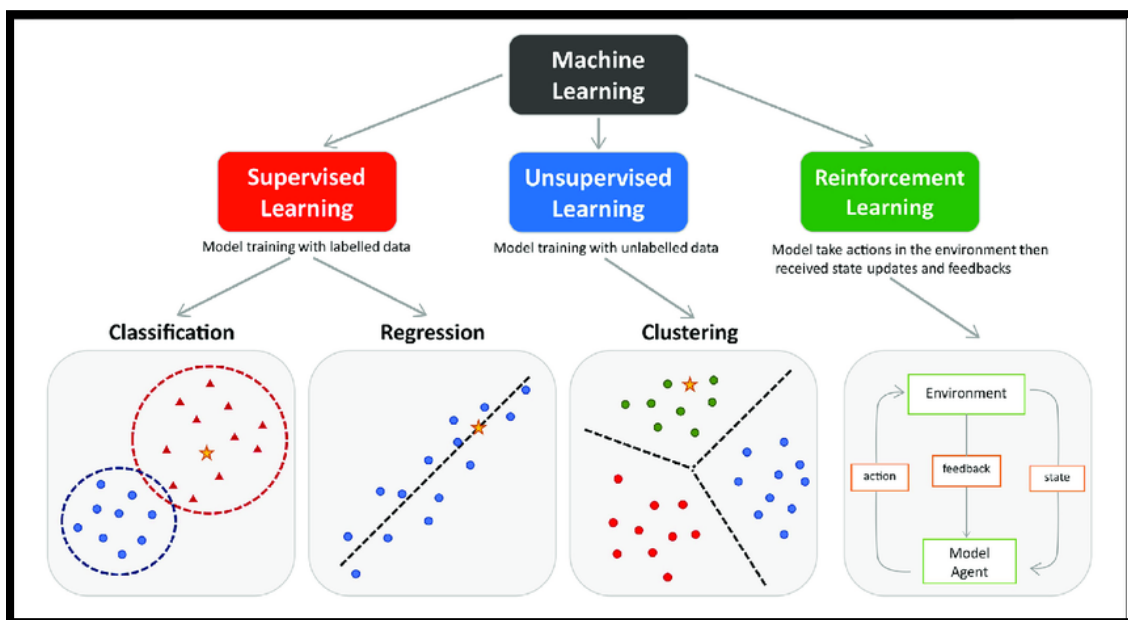
In this article, we will have a detailed analysis on Supervised Learning, exploring its major types and popularly used models.

WHAT IS MACHINE LEARNING?

Machine learning is a subset of Artificial Intelligence . It is the study of statistical and computational techniques which enables a computer to learn from data. The system learns from the data, identifies patterns and generates predictions.

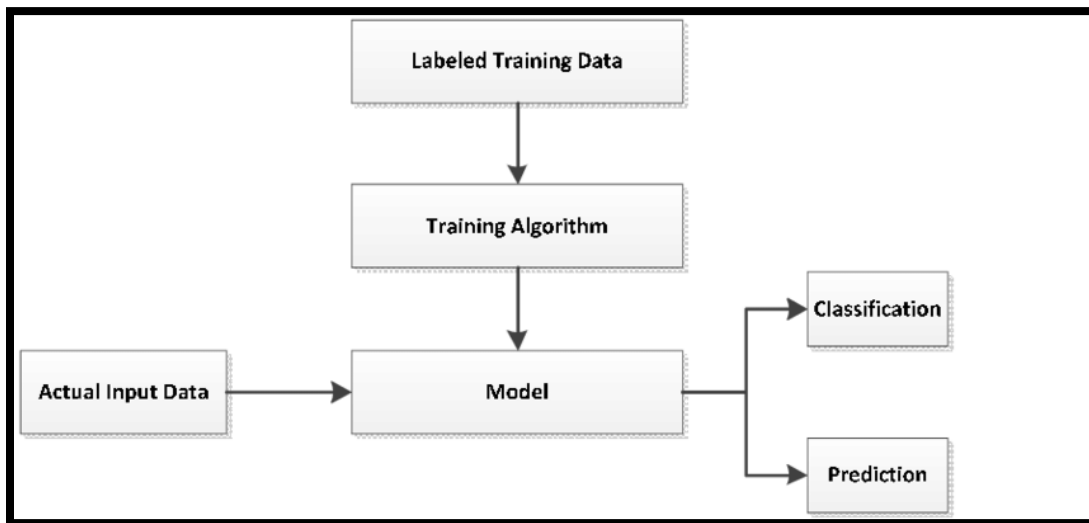
MACHINE LEARNING LEARNING PARADIGMS

Machine Learning is categorised into three broad categories based on the type of data input and learning ability. These categories are then further divided into different classes.



SUPERVISED LEARNING PARADIGM

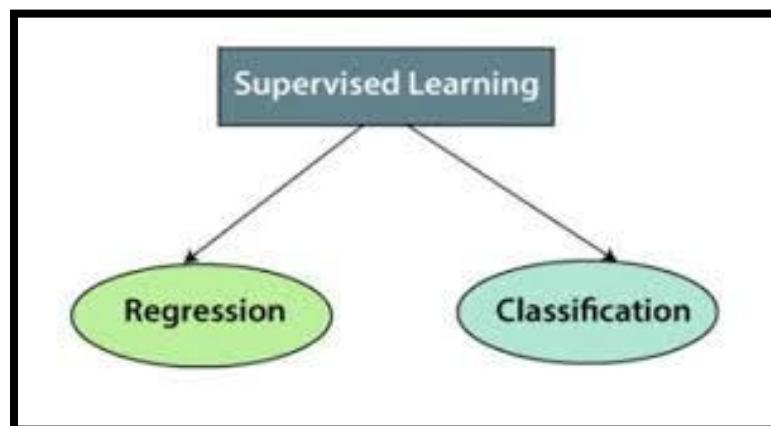
In this type of learning the algorithms are trained with labelled dataset. The labelled dataset contains input variables (features) and their associated outputs (labels). The model learns the relationship between the features and labels and then uses the learned techniques for making predictions .



Block Diagram of Supervised Learning

TYPES OF SUPERVISED LEARNING ALGORITHMS

The two types are as follows:



- **Classification:**

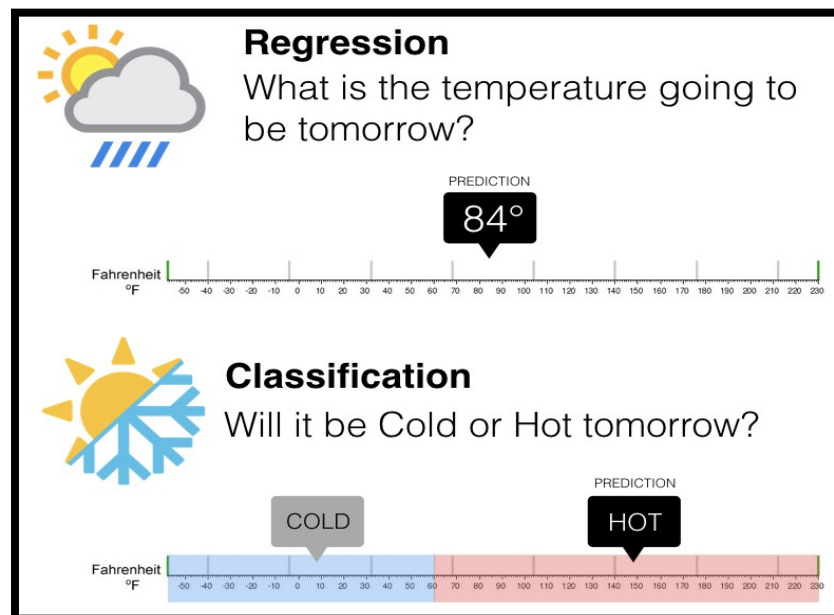
The classification model divides the data into categories and then predicts the probability of a certain type of data falling into the predefined categories. Some examples of this model include :

- filtration of email into 'spam' and 'not spam'.
- Identifying gender as 'Male' or 'Female'
- Identifying a plants specie
- Based on temperature data predict it will 'rain' or 'not rain'

- **Regression:**

The regression model predicts numerical values against the provided input parameters. Some examples of this model include:

- Predicting house prices
- Predicting students marks
- Predicting daily temperature
- Predicting sales



Example of Regression and Classification

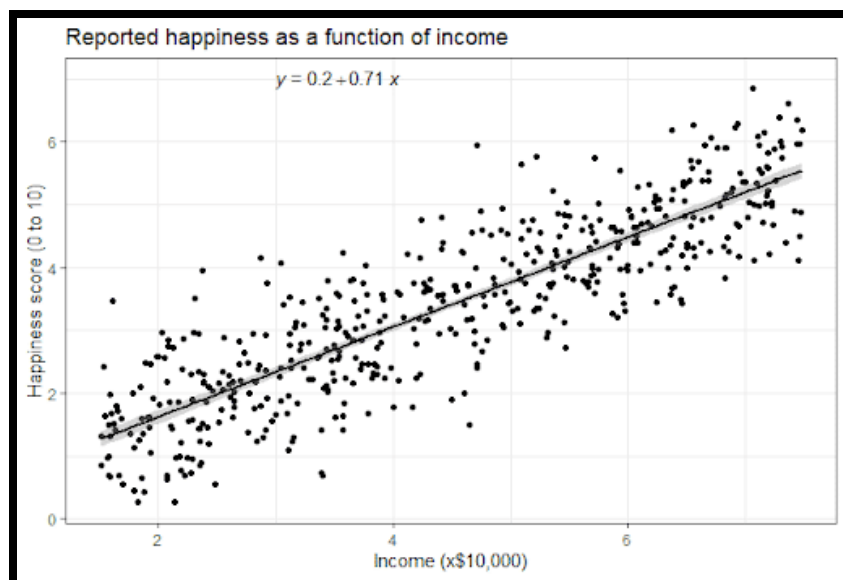
POPULAR REGRESSION AND CLASSIFICATION MODELS

1. Linear Regression

Linear regression is one of the most widely used supervised learning algorithms. which predicts a continuous numeric value . The model is fed with a training dataset and then it is tested upon some unseen data to see how well it has learned to make predictions.

Working:

The model splits the data into 'x' and 'y' variables with 'x' being the independent variable and 'y' being the target or the dependent variable. The target variable can be dependent on one or more variables. The relationship between both types of variables is approximately linear. Mathematically, the linear regression model works on the equation of line $y=mx+c$. It tries to fit a line through the data points to make predictions. The line that fits the best on the data points is known as the **regression line**.



Applications Of Linear Regression:

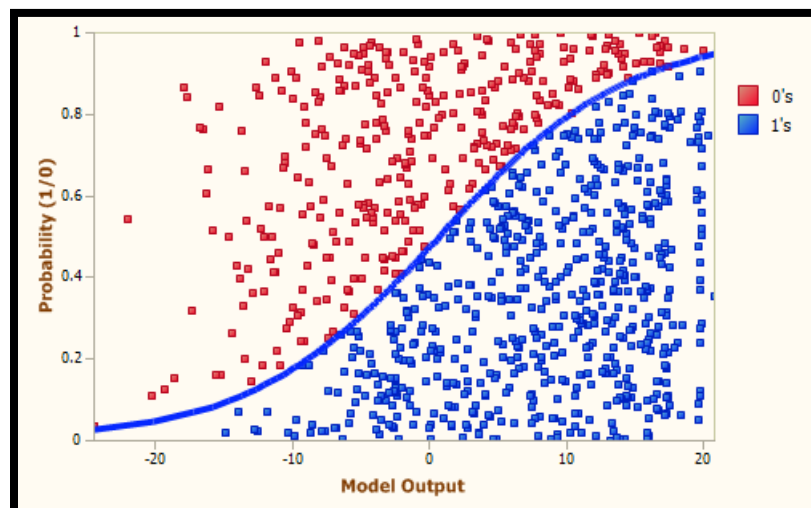
- In marketing, it is used for predicting the impact of advertising expenditure on a product's sales
- In real estate agency, it can be employed in giving house price predictions based on factors like number of room, area , location etc
- In supply chain management, it can be used to estimate the demand for a certain product based on historical sales data.

2. Logistic Regression

The logistic regression model outputs results in a discrete format. It predicts the probability of a data instance belonging to a particular category. It is mostly used in binary classification problems.

Working:

The model analyses the target variable and makes some classes . The target variable is in the form of predefined categories. Each set of independent variables predicts a class. The classes are in the form of 0 or 1 where both represent two different categories.

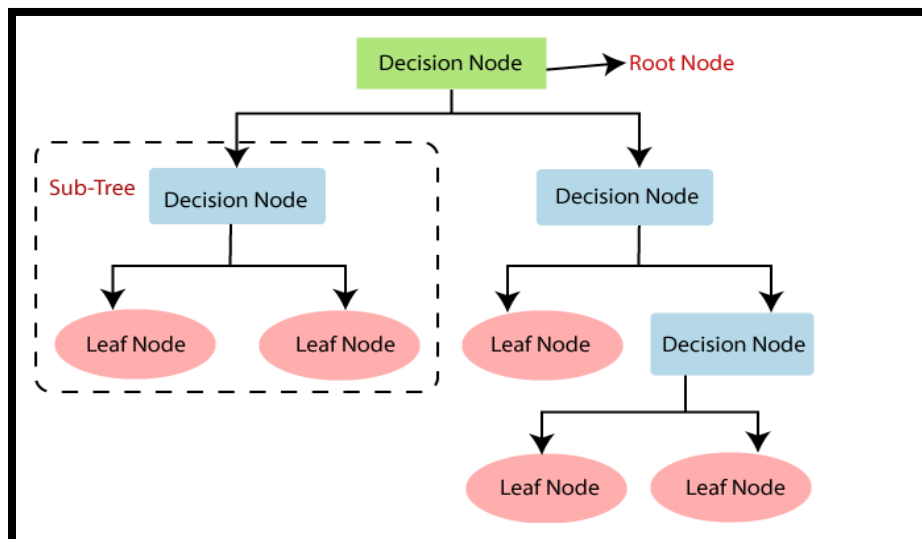


Applications Of Logistic Regression:

- In the healthcare industry, for predicting whether a person suffers from diabetes or not based on factors like age, cholesterol levels, blood pressure, glucose levels etc
- In sports, predicting a team's success or failure. For example, in cricket factors for prediction can be batting score, fielding stats, bowling average etc
- In education, it can be used to analyse whether a student will pass or fail an exam.

3. Decision Trees:

Decision tree is another popular supervised learning algorithm that is used for both classification and regression problems. It is a tree-like data structure which works by recursively splitting the dataset by evaluating the features of the dataset. It is a graphical representation for getting all the possible solutions to a problem.



Working:

The internal nodes of the tree represent the features . These are the decision nodes. On each decision node the tree is further expanded into leaf nodes which represent a particular outcome. The branches of the tree represent the conditions on which a subtree is splitted. A decision evaluates a condition (yes/no) and based on the answer it further splits the tree into subtrees.

For Classification: Decision trees when used for classification problems contain leaf nodes which represent the final decision that is a particular class of the target variable. The decision rules are simple and correspond to all the unique values of features chosen for reaching the final decision.

For Regression: In regression problems , decision trees contain leaf nodes which represent the final decision that is a continuous numerical value of the target variable. The decision rules are simple and correspond to all the unique values of features chosen for reaching the final decision.

Applications Of Decision Trees:

- In fraud detection decision trees are popularly used because of the models ability to handle imbalance data.
- In all of the applications where linear or logistic regression models are applied , decision trees can also be used as they have the tendency to handle both regression and classification tasks.

4. KNN(K - Nearest Neighbours):

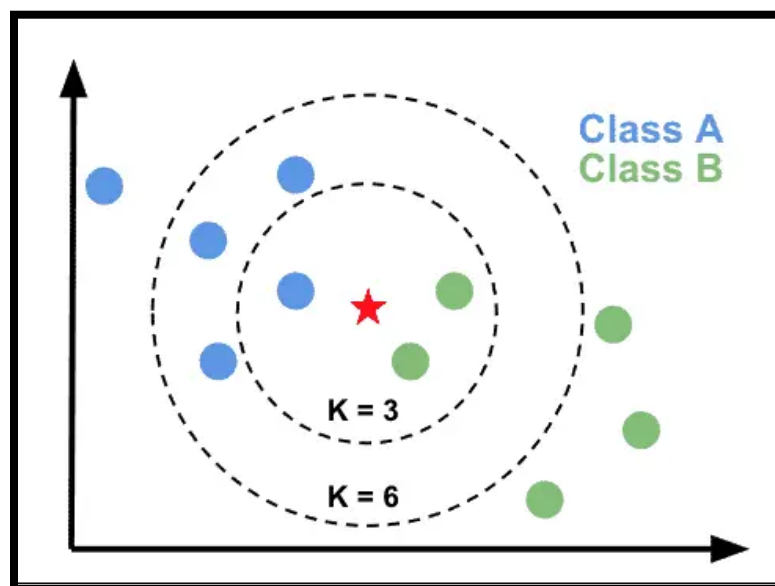
KNN is one of the simplest ML algorithms which can also be used for both classification and regression , but it is mostly used for classification tasks. The KNN algorithm makes predictions by looking for the neighbouring data points (K-Nearest Neighbours) of a particular point and then chooses the majority class. It associates that particular data point to the class of nearest data points.

Working:

The algorithm takes an input parameter 'K' which is the standard of choosing a fixed number of neighbours . This parameter is entered by the user. Then for calculating the nearest proximity , distance metric such as Euclidean Distance is used.

For Classification: KNN works by determining the majority class among the nearest neighbours for a particular data point.

For Regression: For making predictions, KNN determines the average of the nearest neighbours for a particular data point.



Applications Of KNN:

- In image recognition , KNN is used for detecting an object as it can identify patterns in data by comparing it with nearest data points.
- In recommendation systems, KNN is suitable as it gives recommendations based on the preferences of other similar users.

CONCLUSION

In the field of Machine learning , the supervised learning paradigm is evolving rapidly because of its dynamic applications. There are a variety of algorithms offered by this branch of ML and each of them has its own technical capability be it linear or logistic regression, decision trees or KNN or others. But the most important job is the selection of a suitable model.

How Artificial Intelligence Is Transforming Modern Banking

Artificial Intelligence (AI) is rapidly changing the banking industry. From approving loans and detecting fraud to improving customer service and personalizing offers, AI is helping banks operate more efficiently while delivering better experiences to customers.

Traditionally, many banking processes relied heavily on manual work, making them slow, costly, and prone to errors. Today, Machine Learning (ML), Natural Language Processing (NLP), predictive analytics, and Large Language Models (LLMs) are enabling banks to automate operations, make smarter decisions, and reduce risks.

This article explores some of the most impactful applications of AI across lending, payments, and customer service.

AI in Lending and Credit Management

AI-Powered Loan Management

Loan approval is one of the most critical functions of a bank. Traditionally, applications go through multiple checks, verifications, and approval stages, often requiring significant human intervention.

As a result, customers may wait days or even weeks to receive a decision on their loan application.

AI addresses this challenge by automating the entire process. Machine learning models can handle document verification, credit assessment, and decision-making with minimal manual involvement. This significantly reduces processing time and improves the overall customer experience.

Business Impact:

- Faster Turnaround Time (TAT)
- Improved customer satisfaction
- Reduced operational workload

Intelligent Credit Scoring

Before approving a loan, banks must determine whether a customer is likely to repay it.

Traditional credit scoring methods rely heavily on limited data such as income levels and credit history. While useful, these factors often provide only a partial picture of a customer's financial behavior.

AI-powered credit scoring goes beyond traditional methods by analyzing transaction patterns, spending behavior, repayment habits, and other customer data points. This enables banks to assess risk more accurately.

Business Impact:

- Better lending decisions
- Lower default rates
- Faster approvals
- Improved risk management

Automated Document Verification

Banks are required to verify documents such as CNICs, income proofs, and bank statements before approving financial products.

Manual verification is time-consuming and susceptible to human error. It also increases the risk of fraudulent documents slipping through the process.

Using Optical Character Recognition (OCR) and Natural Language Processing (NLP), AI can extract information from documents, validate content, and identify inconsistencies automatically.

Business Impact:

- Faster verification
- Reduced fraud risk
- More secure lending processes

Early Default Prediction

One of the biggest challenges for lenders is identifying customers who may struggle to repay loans.

Traditional systems often detect problems only after customers begin missing payments. By that stage, financial losses may already be significant.

AI uses predictive analytics to monitor repayment behavior and account activity. It can identify warning signs early and alert banks before customers default.

Business Impact:

- Lower Non-Performing Loans (NPLs)
- Better risk control
- Proactive customer management

Collection Optimization

Loan recovery teams often spend significant resources attempting to collect overdue payments.

Traditional collection approaches frequently treat all defaulters similarly, resulting in inefficient use of time and resources.

Machine learning models can prioritize customers based on their likelihood of repayment and recommend the best communication channel and timing for outreach.

Business Impact:

- Higher recovery rates
 - Lower collection costs
 - Improved operational efficiency
-

AI in Payments and Card Services

Smart Payment Advisor

Customers often miss opportunities to maximize rewards, cashback, or discounts because they are unaware of the best payment option available.

AI-powered recommendation engines analyze spending behavior and available promotions to suggest the most beneficial card or payment method during transactions.

Business Impact:

- Increased card usage
- Higher transaction volumes
- Improved customer engagement

Fraud Detection

Fraudulent transactions remain one of the biggest risks in banking.

Traditional rule-based systems may struggle to detect sophisticated fraud patterns quickly enough to prevent losses.

AI analyzes transactions in real time and assigns risk scores based on customer behavior and transaction history. Suspicious transactions can be flagged or blocked immediately.

Business Impact:

- Reduced fraud losses
- Enhanced security
- Greater customer trust

Spend Categorization

Understanding spending behavior is important for both banks and customers.

AI automatically classifies transactions into categories such as groceries, travel, dining, entertainment, and utilities. This provides customers with better visibility into their finances while helping banks deliver personalized services.

Business Impact:

- Better financial insights

- Personalized customer experiences
- Improved financial planning

Offer Personalization

Many banks send the same promotions to all customers regardless of their interests.

This often results in low engagement and poor conversion rates.

AI analyzes customer preferences and spending habits to deliver highly relevant offers tailored to individual behavior.

Business Impact:

- Increased customer engagement
- Higher card usage
- Stronger customer loyalty

Anomaly Detection

Banks continuously monitor transactions for unusual activities that may indicate fraud or security threats.

AI-powered anomaly detection systems identify abnormal transaction patterns in real time and immediately notify the bank or customer.

Business Impact:

- Faster fraud response
 - Reduced financial losses
 - Improved customer confidence
-

AI in Customer Service

AI Chatbots

Customer service departments often experience high volumes of repetitive inquiries.

As a result, customers face long waiting times while support teams become overwhelmed.

AI-powered chatbots built on Large Language Models (LLMs) can instantly answer frequently asked questions and handle basic service requests without human assistance.

Business Impact:

- Lower operational costs
- Faster support
- Improved customer experience

Complaint Management

Banks receive complaints through multiple channels including emails, calls, websites, and mobile applications.

Manually routing complaints to the appropriate department can cause delays and frustration.

Using NLP, AI can analyze complaint content and automatically direct cases to the correct team.

Business Impact:

- Faster issue resolution
- Improved customer satisfaction
- Reduced operational costs

Sentiment Analysis

Understanding customer sentiment is essential for improving services.

Traditional surveys often provide incomplete feedback and may not capture real-time customer emotions.

AI-powered sentiment analysis evaluates customer messages, reviews, emails, and social media interactions to identify satisfaction levels, frustrations, and emerging concerns.

Business Impact:

- Better customer experience insights
- Proactive issue resolution
- Improved service quality

Ticket Routing

Customer support tickets need to be assigned to the right departments quickly.

Manual sorting can slow down response times and increase resolution delays.

Machine learning algorithms automatically classify and route tickets based on their content.

Business Impact:

- Reduced Turnaround Time (TAT)
- Faster resolution
- Greater operational efficiency

Voice Analytics

Every customer service call contains valuable information, yet much of it often goes unused.

Speech analytics technology converts voice conversations into text and analyzes customer emotions, recurring issues, and agent performance.

This enables banks to identify service gaps and improve customer interactions.

Business Impact:

- Better quality control
 - Enhanced employee training
 - Improved customer satisfaction
-

Conclusion

Artificial Intelligence is no longer a future concept in banking—it is becoming a core driver of operational efficiency, risk management, and customer experience. From automating loan approvals and detecting fraud to delivering personalized recommendations and intelligent customer support, AI is transforming how financial institutions operate.

As technology continues to evolve, banks that successfully integrate AI into their processes will be better positioned to reduce costs, improve decision-making, strengthen security, and meet the growing expectations of modern customers. The future of banking is increasingly intelligent, automated, and customer-centric.

1. Problem Identification

Local businesses, particularly small and medium-sized enterprises (SMEs), are increasingly in need of affordable AI chatbot solutions to enhance customer engagement and automate support processes. However, most advanced chatbot platforms available today are priced in dollars, making them financially out of reach for many businesses operating on tight budgets. Moreover, in Pakistan where Urdu is the national language and widely preferred for communication, existing chatbots often fail to effectively cater to Urdu-speaking users. This situation leaves a significant gap for an accessible and budget-friendly chatbot service tailored to local needs.

2. Significance

Our product aims to fill this gap by offering a cost-effective, bilingual chatbot service that supports both English and Urdu, enabling local businesses to deliver customer service without the financial burden of expensive solutions. This product would be beneficial for organizations and services operating in regions where both English and Urdu are spoken, improving accessibility and user experience.

3. Objectives

- Develop a cost-effective AI-based customer engagement tool that is accessible to businesses with limited budgets.
- Enable seamless communication in both English and Urdu to cater to the linguistic preferences of a diverse customer base.
- Improve customer experience by offering a solution that bridges the gap between existing expensive chatbot platforms and the needs of local businesses.
- Empower organizations to deliver automated, high-quality customer support without significant financial burdens.

4. Project Scope

Our project has two key components:

Bilingual Communication Support

This component focuses on enabling businesses to create chatbots that can interact seamlessly in both English and Urdu, with an initial greeting that allows the user to choose their preferred language for communication.

Customizable Chatbot Creation and Integration

The product will be a website platform where businesses can easily design and create chatbots. These chatbots can be customized to fit specific needs and then integrated into the business's own website for customer engagement.

Target Population

Our target population is mainly small-scale businesses and organizations. These businesses will be able to create and deploy chatbots that communicate in both English and Urdu, improving their customer service capabilities.

Geographical Scope

Our project is designed for businesses operating in regions where both English and Urdu are spoken, with a primary focus on Pakistani businesses. The solution is adaptable for other multilingual markets as well.

Technological Scope

Our project encompasses the following technological aspects:

Multilingual NLP Integration

Leveraging Large Language Models (LLMs) to provide accurate, context-aware responses in both English and Urdu. Users will have the option to select their preferred language (English or Urdu) at the start of the interaction, with the chatbot adapting the conversation accordingly.

Web-Based Platform for Chatbot Creation

A user-friendly website where customers, such as businesses and organizations, can design and build their chatbots. The platform facilitates the seamless setup of multilingual chatbots, allowing them to cater to diverse user bases.

Chatbot Deployment and Website Integration

Once the chatbot is created, businesses can easily integrate it into their existing websites. The platform generates a JavaScript code that can be embedded into the website, enabling direct customer interactions.

FAQ and Customer Support Features

The chatbots will include built-in capabilities for handling frequently asked questions (FAQs), responding to customer inquiries, and providing basic assistance. These features will be fully functional in both English and Urdu, depending on the user's language preference.

Frontend

The frontend will be the interface where business owners can interact with your platform. It's essentially the website where they'll upload their documents/URLs and configure the chatbot.

Key Features of the Frontend

User Dashboard

A simple and intuitive dashboard where users can:

- Upload a PDF or provide a website URL.
- See and manage their existing chatbots.

File/URL Upload Interface

A form where users can either paste a URL or upload a PDF document.

Chatbot Configuration Options

Users can set preferences, such as:

- Name and greeting message for the chatbot in both English and Urdu.
- Select predefined FAQ formats or provide custom questions.

Integration Code Generator

Once the chatbot is created, provide a JavaScript code snippet that users can embed into their own websites to deploy the chatbot.

Preview and Testing

Allow users to interact with the chatbot directly from your platform before integrating it into their website.

Tech Stack for Frontend

- Frameworks/Libraries: React.js (for building dynamic UIs)
- Styling: Tailwind CSS, Bootstrap, or Material UI
- File Handling: Libraries like FilePond or Dropzone for handling file uploads
- API Integration: Axios or Fetch API for connecting to your backend

Backend

The backend will handle the actual processing, from content extraction and chatbot creation to serving responses.

Key Components of the Backend

Content Extraction Service

Document/URL Loader

Use LangChain's document loaders for PDF and web content extraction.

- For URLs: Use WebPageLoader.

- For PDFs: Use PyPDFLoader or similar document loaders.

Text Processing

Use text splitters to break down large content into manageable chunks for indexing.

LLM Fine-Tuning and Chatbot Logic

Bilingual Model Handling

Fine-tune or use pre-trained bilingual models (like mT5 or XLM-R) that handle both English and Urdu.

RAG Pipeline Integration

Implement a Retrieval-Augmented Generation (RAG) pipeline using LangChain.

- Store extracted content in a vector database (like FAISS) and use it to generate responses.

Dynamic Indexing

When a business uploads new documents or provides a new URL, index the content dynamically and make it available to the chatbot.

API Development

Create RESTful or GraphQL APIs that:

- Accept file uploads or URLs.
- Configure chatbot settings (like languages).
- Serve the chatbot responses when end-users interact with the chatbot.

Database Management

Store user details, chatbot configurations, and content indexing metadata.

- Use relational databases like PostgreSQL or NoSQL solutions like MongoDB for flexibility.

Authentication and Security

- Implement user authentication (e.g., using JWT).
- Secure file uploads and sanitize inputs to prevent malicious content.

Chatbot Deployment and Integration

- Generate JavaScript code for embedding the chatbot in the business's website.
- Host the chatbot backend with endpoints that serve real-time responses when users interact.

Tech Stack for Backend

- Programming Language: Python Flask or Node.js (Express)
- Web Framework: FastAPI (Python) for speed and efficiency
- LLM Integration: LangChain for managing LLMs, embeddings, and retrieval pipelines
- Vector Database: FAISS or Pinecone for storing and retrieving document embeddings

- APIs: FastAPI for creating endpoints
- Database: MongoDB
- Cloud Hosting: AWS, GCP, or Heroku for scalable deployments

Overall Workflow

Business Owner Interaction

The business owner logs in, uploads a document or provides a URL, and configures their chatbot.

Backend Processing

- The backend extracts the content from the URL/PDF.
- The content is preprocessed, split into chunks, and embedded using an LLM.
- The embeddings are stored in a vector database, ready for retrieval during user queries.

Chatbot Creation and Integration

The backend generates a JavaScript code snippet for the business owner to integrate into their website.

End-User Interaction

When the end-user visits the business's website and interacts with the chatbot, their queries are processed using the LLM and the RAG pipeline, fetching relevant information from the vector database based on the uploaded content.

5. Methodology Overview

Our development approach focuses on developing a website where local businesses can easily generate bilingual chatbots for their websites. The project is divided into well-defined phases to help us in delivering a solution that caters to the specific needs of local businesses.

Project Workflow

Project Planning and Requirements Gathering

The project would begin with comprehensive planning and gathering of all necessary requirements, focusing on user needs and product features.

Front-End Development

The front-end of the platform will be developed with an emphasis on a user-friendly interface, allowing businesses to easily create and customize chatbots.

Back-End Development

The back-end will focus on building the core functionalities that drive the chatbot creation process. This includes server-side logic, database integration, and ensuring the system can handle real-time interactions.

AI Model Integration

In this phase, we will integrate AI models that support bilingual communication (English and Urdu). These models will be fine-tuned to provide accurate and context-aware responses based on user input and the uploaded content (URLs or documents). The RAG pipeline would also be introduced.

Real-Time Communication Implementation

This phase will involve implementing the real-time communication capabilities of the chatbots.

Testing and Quality Assurance

Comprehensive testing will be conducted to ensure the chatbots perform as expected, providing accurate responses in both languages and handling various customer queries effectively.

Deployment and Monitoring

The platform and chatbots will be deployed for real-world use, followed by continuous monitoring to track performance and user feedback.

This structured approach would help us in keeping the project timeline on track while allowing for adaptive iterations during key development phases.

Resources Involved

- Figma: For designing user interfaces and prototypes.
- React JS: For building a dynamic and responsive front-end interface.
- Bootstrap: To streamline front-end development with pre-designed components.
- Node JS: For developing a scalable and efficient backend system.
- Large Language Models (LLMs): Leveraging models like GPT-3.5/4 for multilingual chatbot capabilities, fine-tuned to handle both English and Urdu content.
- LangChain: To manage LLM-powered tasks such as content extraction from documents and generating conversational responses.
- Microsoft Azure: For cloud-based deployment, hosting, and AI model integration.
- MongoDB: For handling database management, including storing user data, chat histories, and chatbot configurations.

6. Conclusion

This project aims to address the growing need for affordable and accessible AI-powered chatbot solutions for local businesses. By providing bilingual support in both English and Urdu, the platform seeks to bridge the gap between expensive international chatbot services and the specific requirements of businesses operating in multilingual environments. Through the integration of Large Language Models, Retrieval-Augmented Generation (RAG), and a user-friendly chatbot creation platform, the proposed solution enables organizations to enhance customer engagement, automate support services, and improve accessibility. The project ultimately focuses on delivering a practical, scalable, and cost-effective solution tailored to the needs of the local market.

PRODUCT REPORT

2024 AI CHALLENGE- STUTTERING THERAPY SYSTEM

PRODUCT

- **Problem**

Stuttering is a speech disorder characterized by interruptions in the flow of speech. This speech disorder is mainly divided into three types; repetitions, prolongations, and blocks. Stuttering affects individuals of all ages and can have a significant impact on their communication abilities and self-confidence.

- **Current Solutions**

Current speech therapy methods primarily target enhancing speech fluency through techniques like breathing exercises and deliberate speech. However, these approaches can be inaccessible and costly. Hence, there's a demand for an innovative, personalized solution to better aid individuals dealing with stuttering.

- **Our Solution**

Our system offers a comprehensive approach to address this issue. Here's a brief overview of the functionalities:

1. **Stuttering Detection:** Utilizing cutting-edge speech analysis algorithms, our system accurately detects instances of stuttering in individuals' speech patterns.
2. **Classification into Main Types:** Identified instances of stuttering are classified into main types such as prolongation, repetition, and blocks.
3. **Personalized Therapy Provision:** For each identified type of stuttering, therapy resources in the form of audio recordings are provided.

- **Target Audience**

Our system is for individuals of all ages facing stuttering or speech fluency issues, with a specific emphasis on those who stutter, regardless of age. Additionally, speech-language therapists and professionals assisting individuals with stuttering can also find value in our system.

ENGINEERING

High-Level Design:

Our stuttering detection system operates on a two-fold approach: Audio classification and Therapy generation. Using the DistilHubert transformer model, audio inputs are classified into distinct stuttering types. This process involved preprocessing the data, training the model on our own dataset, and optimizing its performance through fine-tuning. The system's architecture ensures efficient classification of stuttering types, providing users with valuable insights into their speech patterns.

Role of Generative AI (MMS Model Text-to-Speech Therapy Recommendation):

We used the Massively Multilingual Speech (MMS) English Text-to-Speech model for therapy generation based on the identified stuttering type. By using it as an API in our system, users can receive therapy in the form of audios.

Technologies Used:

- DistilHubert (transformer model)
- Massively Multilingual Speech (MMS) (text-to-speech model)
- Hugging Face Transformers library
- Gradio library
- TensorFlow
- PyTorch

RESULTS AND ANALYSIS

In this section, we present the results of our stuttering detection model which was fine tuned on the **Distil Hubert** model using our own dataset.

- Evaluation Approach: For evaluating the model's performance, we computed various metrics, including accuracy, on an independent test set.
- Loss: The model achieved a loss value of 0.5717, indicating the variation between the predicted values and the ground truth labels during training.
- Accuracy: With an accuracy score of 0.9024, our model correctly classified approximately 90.24% of the samples in the evaluation set.

FUTURE PROSPECTS

1. Multimodal Inputs: Integrate additional features such as facial expressions or physiological signals to improve the accuracy and robustness of stutter detection.
2. Personalized Therapy: Tailor therapy recommendations to individual preferences and needs.
3. Real-Time Feedback: Provide immediate support during speech interactions for improved fluency.
4. Accessibility and Scalability: Optimize resources and support multiple languages for broader user reach.
5. Collaboration with Healthcare Professionals: Partner with experts to validate effectiveness and ensure alignment with established practices.